F78AA - Actuarial and Financial Mathematics A
Tutorial 4 – Loan schedules

Question 1 You borrow £50 000 for 4 years from a building society to buy a flat. The loan is to be repaid by level annual repayments (note: this always means payment in arrears, even if not stated). The interest rate is 6% per annum effective.

Calculate the annual repayment and draw up the complete loan schedule.

Question 2 Anna borrows £50 000 from Bob, for a term of 25 years. The loan is to be repaid by level annual repayments, and the interest rate is 4.5% per annum effective. Calculate:

1. the annual repayment.
2. the capital outstanding immediately after the 5th repayment.
3. the capital content and interest content of the 20th repayment.

Question 3 Anna borrows £30 000 from Bob, for a term of 15 years. The loan is to be repaid by level monthly repayments, and the interest rate is 7% per annum effective. Calculate:

1. the monthly repayment.
2. the capital outstanding immediately after the 100th repayment.
3. the capital content and interest content of the 111th repayment.

Question 4 You borrow £35 000 from a bank, to be repaid over 18 years by level annual repayments. The interest rate is 6.5% per annum effective. Immediately after the 10th repayment has been made, you experience financial difficulties. You ask the bank if you may:

(a) miss the next two repayments, or

(b) extend the term of the loan by an extra 3 years.

1. Calculate the revised annual repayments that the bank would ask you to make in each case.
2. Suppose that you wanted to miss the next two repayments *and* extend the term of the loan by an extra 3 years. Calculate the new annual repayment.
3. Suppose the bank agrees to let you miss the next two repayments, but as a penalty, the capital outstanding will be charged interest at 9% per annum while you are not making repayments. Calculate the new annual repayment.

Question 5 Assume the set-up of Q4.

1. In each case considered in Q4.1, write down the equation of value at the end of the loan period and then verify that the yield on the transaction is still 6.5% per annum.
2. In the case considered in Q4.3, write down the equation of value at the end of the loan period and then calculate the yield on the transaction.

Question 6 You borrow £80 000 from a building society. The term of the loan is 20 years and you have to repay it by level annual repayments. The interest rate will be 5% per annum effective for the first 5 years, and 6% per annum effective thereafter.

1. Calculate the annual repayment.
2. Immediately after the 7th repayment has been made, the building society raises the rate of interest for all borrowers from 6% per annum to 8% per annum effective.

   Calculate the new repayment.

Question 7 A bank charges interest of 8% per annum on all loans. However, as a special incentive to new borrowers it will waive all interest on a loan during the first two years.

You take out a loan of £10 000 for a term of 10 years, repayable by level annual repayments. Calculate the annual repayment.

Question 8 Anna borrows £20 000 from Bob for a term of 15 years. The interest rate will be 10% per annum effective for the first 10 years, and 1% per month effective thereafter. The loan is to be repaid by annual repayments which will be level for the first 10 years, then increasing to twice the original level for the last 5 years.

1. Calculate the level of the repayment for the first 10 years.
2. Immediately after the 6th repayment has been made, Anna requests that the future repayments should be monthly instead of annual (still doubling at time 10 years). Calculate the new monthly repayment until time 10 years.

Question 9 You borrow £25 000 from a bank, to be repaid over a 20-year term. Repayments are yearly in arrears, increasing at 3% per annum. The rate of interest is 6% per annum effective. Calculate:

1. the first repayment.
2. the capital outstanding just after the 10th repayment is made.
3. the interest content of the 11th repayment.

Solutions: Q1. Repayment £14 429.57; Q2.1. £3 371.95; Q2.2. £43 862.11; Q2.3. Interest £782.64, capital £2 589.31; Q3.1. £266.05; Q3.2. £17 082.84; Q3.3. Interest £86.76, capital £179.29; Q4.1. Case (a) £4 785.99, Case (b) £2 656.67; Q4.2. £3 480.87; Q4.3. £5 013.32; Q5.2. 0.0668; Q6.1. £6 700.57; Q6.2. £7 505.02; Q7. £1 290.88; Q8.1. £2 252.81; Q8.2. £178.42; Q9.1. £1 716.85; Q9.2. £19 194.00; Q9.3. £1 151.64.

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Solutions to Tutorial 4 – Loan schedules

Solution 1 The repayment is $50\,000/a_{\overline{4}|}^{0.06} = £14\,429.57$.

The loan schedule is:

Time	Repayment	Interest Content	Capital Content	Capital Outstanding
0				50 000.00
1	14 429.57	3 000.00	11 429.57	38 570.43
2	14 429.57	2 314.23	12 115.34	26 455.09
3	14 429.57	1 587.31	12 842.26	13 612.83
4	14 429.57	816.77	13 612.80	0.03

In practice, we would make three payments of £14 429.57 and a final payment of £14 429.60.

Solution 2

1. $X = 50\,000/a_{\overline{25}|}^{0.045} = £3\,371.95$
2. $Xa_{\overline{20}|}^{0.045} = £43\,862.11$.
3. The capital outstanding just after the 19th repayment is $Xa_{\overline{6}|}^{0.045} = £17\,392.09$. So the interest content of the 20th repayment is $0.045 \times 17\,392.09 = £782.64$. The capital content of the 20th repayment is $X - 782.64 = £2\,589.31$.

Solution 3 Treat the loan as one of 180 months at an effective rate of interest of 0.005654 per month.

1. $X = 30\,000/a_{\overline{180}|}^{0.005654} = £266.05$.
2. The capital outstanding just after the 100th repayment is $Xa_{\overline{80}|}^{0.005654} = £17\,082.84$.
3. The capital outstanding just after the 110th repayment is $Xa_{\overline{70}|}^{0.005654} = £15\,344.43$. So the interest content of the 111th repayment is $0.005654 \times 15\,344.43 = £86.76$. The capital content of the 111th repayment is $X - 86.76 = £179.29$.

Solution 4

1. The original repayment is $X^{(0)} = 35\,000/a_{\overline{18}|}^{0.065} = £3\,354.91$. The capital outstanding just after the 10th repayment is $X^{(0)} \times a_{\overline{8}|}^{0.065} = £20\,427.21$.

   Case (a): In two years, with interest at 6.5% p.a, this accumulates to £23 169.05. This must now be repaid by only 6 annual repayments, so the new repayment is $X^{(1)} = 23\,169.05/a_{\overline{6}|}^{0.065} = £4\,785.99$.

   Case (b): The new repayment is $X^{(2)} = 20\,427.21/a_{\overline{11}|}^{0.065} = £2\,656.67$.
2. The new repayment would be $X^{(3)} = 23\,169.05/a_{\overline{9}|}^{0.065} = £3\,480.87$.
3. The capital outstanding would accumulate to $20\,427.21 \times 1.09^2 = £24\,269.57$, so the new repayment would be $X^{(4)} = 24\,269.57/a_{\overline{6}|}^{0.065} = £5\,013.32$.

Solution 5

1. From the previous question, the original repayment is $X^{(0)} = £3\,354.91$.

   Case (a): From the previous question, the new repayment is $X^{(1)} = £4\,785.99$. Hence the equation of value is:

   $$f(i) = 35\,000\,(1+i)^{18} - 3\,354.91\,s^{i}_{\overline{10}|}(1+i)^{8} - 4\,785.99\,s^{i}_{\overline{6}|} = 0.$$

   It is verifiable by direct substitution in to the equation of value that $i^\star = 0.065$ is a solution of this.

   Case (b): From the previous question, the new repayment is $X^{(2)} = £2\,656.67$. Hence the equation of value is:

   $$f(i) = 35\,000\,(1+i)^{21} - 3\,354.91\,s^{i}_{\overline{10}|}(1+i)^{11} - 2\,656.67\,s^{i}_{\overline{11}|} = 0.$$

   It is verifiable by direct substitution in to the equation of value that $i^\star = 0.065$ is a solution of this.

2. From the previous question, the new repayment is $X^{(4)} = £5\,013.32$.

   Hence the equation of value is:

   $$f(i) = 35\,000\,(1+i)^{18} - 3\,354.91\,s^{i}_{\overline{10}|}(1+i)^{8} - 5\,013.32\,s^{i}_{\overline{6}|} = 0.$$

   As we cannot solve the equation of value exactly, we use trial-and-error to find an approximate solution. We know that the yield must be greater than 6.5% p.a. because the lender is receiving larger repayments.

   As $f(0.065) = -1\,605.74$ and $f(0.07) = +2\,792.91$, and we seek $i^\star$ such that $f(i^\star) = 0$,

   $$\frac{i^\star - 0.065}{0.07 - 0.065} = \frac{f(i^\star) - f(0.065)}{f(0.07) - f(0.065)} \quad \Rightarrow \quad i^\star \approx 0.065 + 0.005\,\frac{1\,605.74}{1\,605.74 + 2\,792.91} = 0.0668.$$

Solution 6

1. Applying the Principle of Equivalence,

   $$80\,000 = X\,(a^{0.05}_{\overline{5}|} + 1.05^{-5}\,a^{0.06}_{\overline{15}|}) \qquad \Rightarrow \qquad X = £6\,700.57.$$

2. The capital outstanding is the present value of the remaining 13 repayments, which is $X \times a^{0.06}_{\overline{13}|} = £59\,318.02$. Therefore the new repayment is $X' = 59\,318.02/a^{0.08}_{\overline{13}|} = £7\,505.02$.

Solution 7 Denote by X the annual repayment. Since no interest accrues on the loan for the first two years, the interest content of the first two repayments is zero, and the capital outstanding after 2 years is $£(10\,000 - 2X)$. This must be repaid over 8 years by level annual repayments of X. By the Principle of Equivalence, after 2 years,

$$10\,000 - 2X = X\,a^{0.08}_{\overline{8}|} \qquad \Rightarrow \qquad X = \frac{10\,000}{2 + a^{0.08}_{\overline{8}|}} = £1\,290.88.$$

Solution 8

1. Let the original level of repayment be X. Applying the Principle of Equivalence,

$$20\,000 = X\,(a_{\overline{10}|}^{0.10} + 2 \times 1.10^{-10} \times a_{\overline{5}|}^{0.126825}) \qquad \Rightarrow \qquad X = £2\,252.81.$$

2. The capital outstanding is the present value of the future repayments,

$$X\,(a_{\overline{4}|}^{0.10} + 2 \times 1.10^{-4} \times a_{\overline{5}|}^{0.126825}) = £18\,049.41.$$

   The effective rate of interest is 0.007974 per month for the next 48 months, and 0.01 per month for the last 60 months, so the new monthly repayment Y is

$$18\,049.41 = Y\left(a_{\overline{48}|}^{0.007974} + 2 \times 1.007974^{-48} \times a_{\overline{60}|}^{0.01}\right) \qquad \Rightarrow \qquad Y = £178.42.$$

Solution 9

1. Set $1 + i' = \frac{1.06}{1.03} = 1.029126$. By the Principle of Equivalence,

$$25\,000 = X \times 1.03^{-1} \times a_{\overline{20}|}^{i'} \qquad \Rightarrow \qquad X = £1\,716.85.$$

2. The amount of the 11th repayment is $X \times 1.03^{10} = £2\,307.30$. So the capital outstanding just after the 10th repayment is made, which is the present value of the remaining 10 repayments, is:

$$2\,307.30 \times 1.03^{-1} \times a_{\overline{10}|}^{i'} = £19\,194.00.$$

3. The interest content of the 11th repayment is $0.06 \times 19\,194.00 = £1\,151.64$.